



Water Wells Functionality in Tanzania

Kuria's Company



Project Overview

This project aims to analyze Tanzanian well data and formulate models that aid the Ministry of Water and Donors in Construction, Maintenance and Repair of wells across the country.

Different models are utilized and the best will be used to make predictions and guide the conclusions made.



Objectives

1:

- Identify factors influencing well functionality

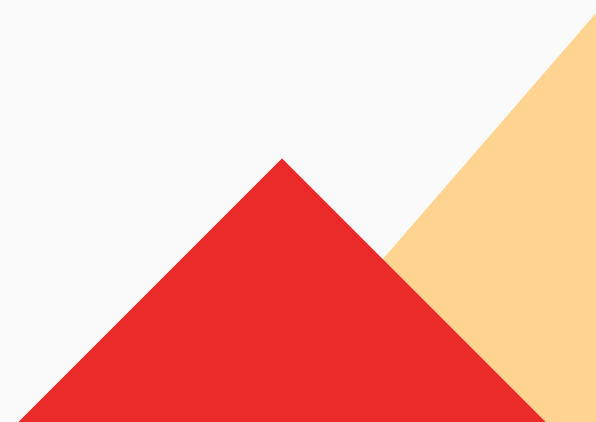
3:

- Recommend well repair strategies

2:

- Detect patterns of non functionality in wells.

4:

- Formulate draft plans for new well construction.
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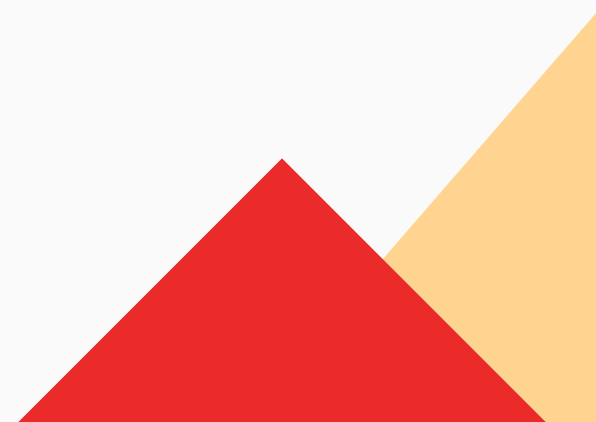


Data Understanding

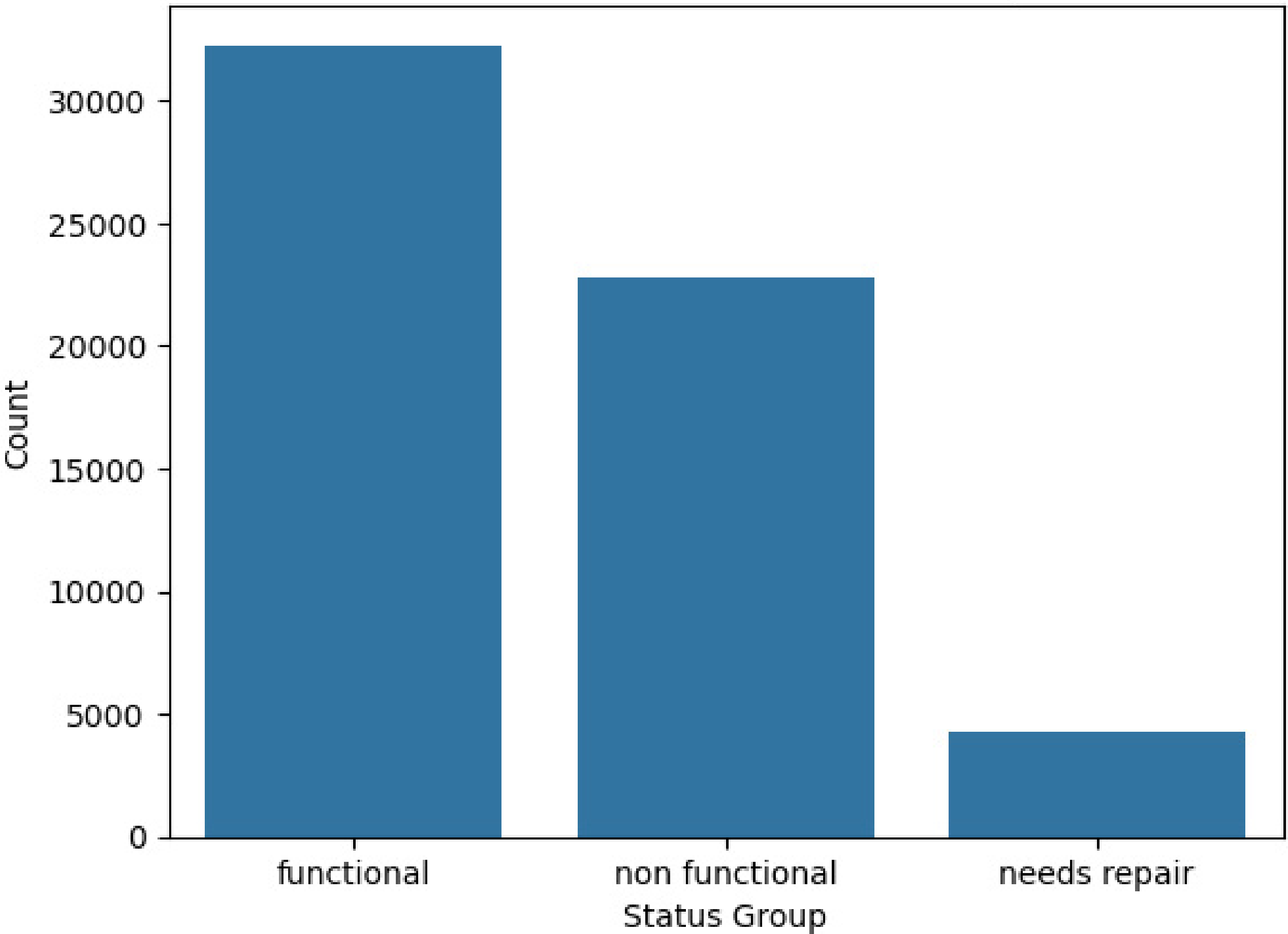
Training and Testing sets of Tanzanian Water Wells were utilized in this project.

Select Feature columns such as location and water quality were utilized .

The Target was the Well Functionality(Status group).

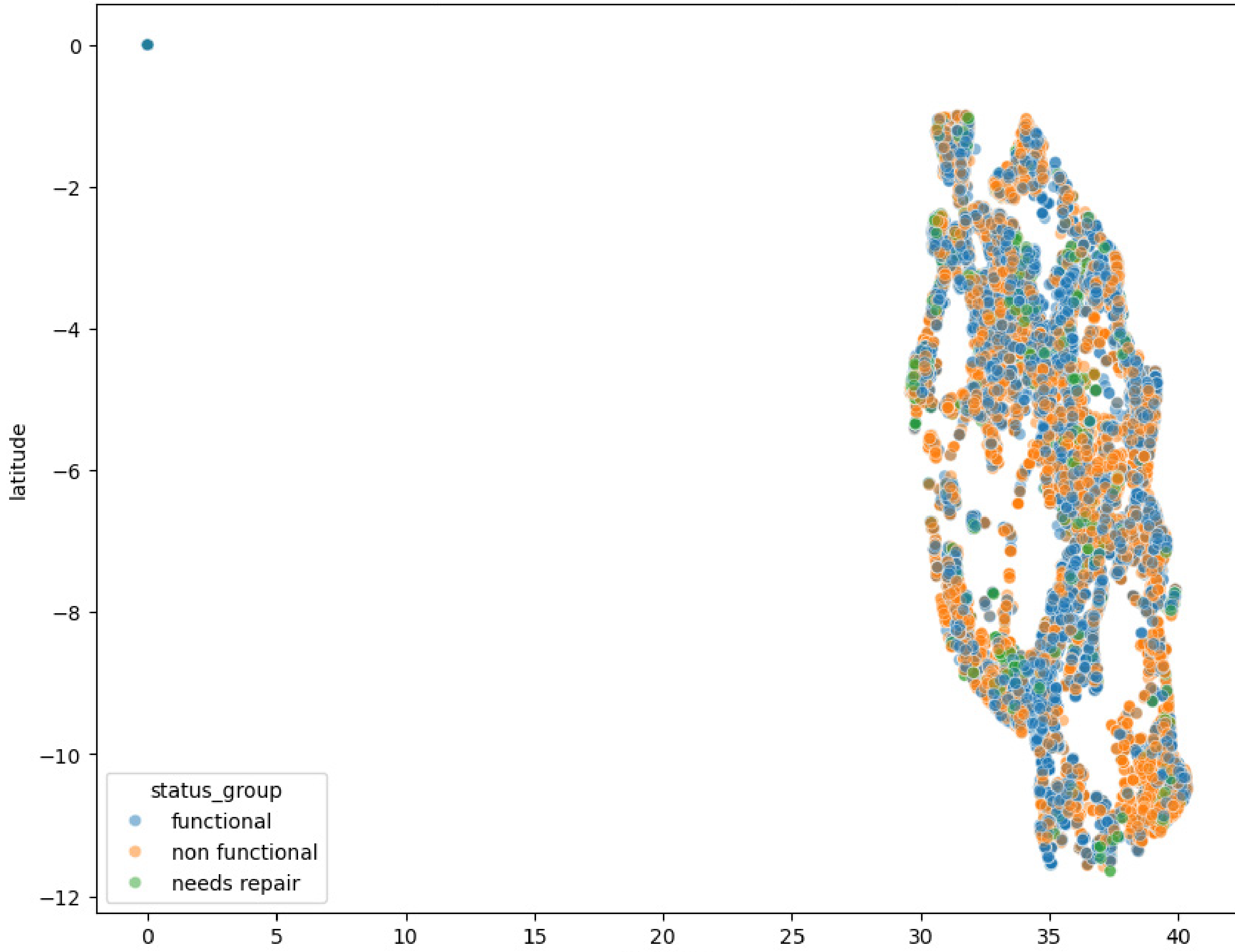


Distribution of Well Functionality



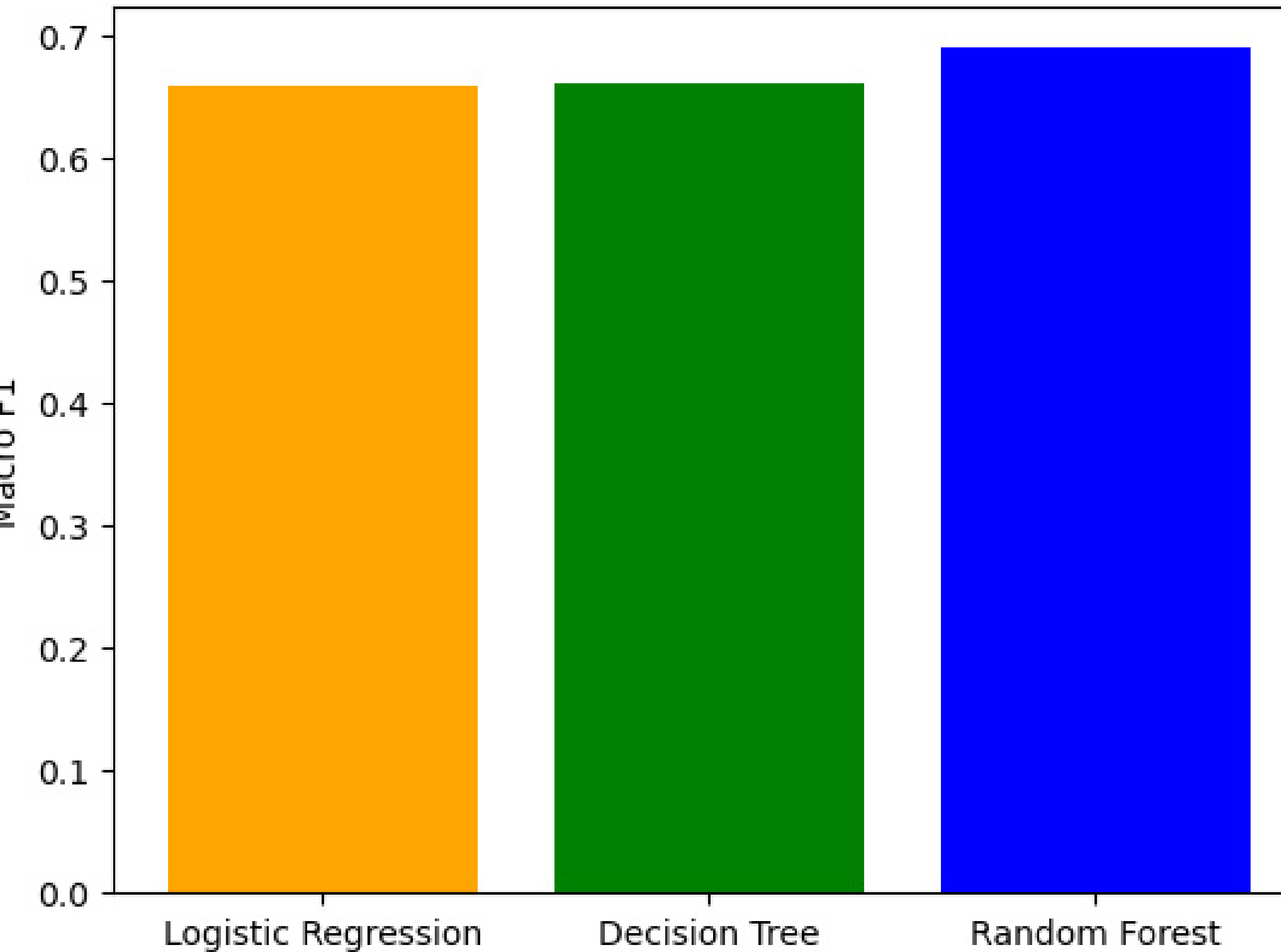
- This bar graph shows the distribution of wells functionality in the country.
- Only a few number of wells require reapi, but a huge number are non functional. This shows the importance in construction and maintenance of the wells in order to utilize full usage for extended periods of time.

Geospatial Distribution of Wells by Functionality



- This shows the distribution of the wells according to location in the country, indicating the locations with the most and least number of wells.

Model Comparison: Macro F1 Scores



- The model comparison shows that the combined decision trees worked best thus were utilized.



Recommendations

- 1. New wells should be constructed in low-risk areas with reliable installers to reduce the need for frequent repairs.**
 - 2. Highly functional water sources should be utilized and constructed more.**
 - 3. Introduce manageable fees to the community that will source income for well repairs.**
 - 4. Constant management and monitoring of the water sources so as to track reliability and factors affecting it.**
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Conclusion

Conclusively, the Random forest model shows that strategic locations should be picked for well construction.

With deployment of the model and regular data training from new data, the model can provide more insights into new constructions, repairs and in showing the features affecting the wells functionality the most.

